An Accelerated Solution Method for Stochastic Optimal Power Flow

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***Abstract*—The large-scale integration of renewable energy introduces significant uncertainty challenges to the optimal dispatch of power systems. To fully account for the impact of the randomness in renewable generation, this paper proposes an efficient solution algorithm based on the Gaussian mixture model (GMM) and the unscented transformation (UT) method, aimed at solving the stochastic optimal power flow (S-OPF) problem. Specifically, this paper first constructs the S-OPF model that considers a series of chance constraints. Next, the GMM is used to fit random variables with arbitrary distributions and the UT algorithm approximates the probabilistic characteristics of input variables with a small number of sample points. A Gaussian component simplification strategy is then proposed to minimize the computational burden while retaining the maximum reconstruction accuracy. Finally, the GMM-UT-based output variable probabilistic reconstruction algorithm is embedded into a general solution framework. Case studies demonstrate the advantages of the proposed algorithm over existing approximation methods and its higher computational efficiency compared to the traditional Monte-Carlo method.**

***Index Terms*—Stochastic optimal power flow, chance constraint, Gaussian mixture model, unscented transformation, Gaussian component simplification.**

I. Introduction

In RECENT years, due to the dual challenges of increasing pressure on power supply and environmental pollution, non-fossil energy sources such as wind and solar energy are being rapidly developed and integrated [1]. However, large-scale renewable energy sources, such as photovoltaics and wind power, face the issue of insufficient energy supply stability due to their inherent volatility and uncertainty characteristics. In the face of the challenges posed by the random variations of numerous uncertain factors in the power system, the traditional deterministic optimal power flow (D-OPF) model exhibits certain limitations [2].

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Scholars have proposed various methods to incorporate randomness into the D-OPF model. One widely studied technique is stochastic optimal power flow (S-OPF), which extends the traditional D-OPF model by introducing chance constraints. These constraints allow the dispatch solution to violate safety constraints under certain conditions, but the violation probability must not exceed a pre-set confidence level. However, in the S-OPF problem, the tight coupling of chance constraints with power flow equations significantly increases the difficulty of solving the problem [3].

In response to the challenges encountered in solving the S-OPF, there are two main solution frameworks: single-step solving and iterative solving. The essence of the single-step solving strategy is to transform chance constraints into deterministic constraints, allowing direct solution using solvers. Vrakopoulou et al. [4], [5] calculated the minimum number of scenarios required to meet a given violation probability, ensuring these scenarios satisfied the constraints to guarantee that the violation probability is met. However, the scenario method comes with a high computational burden and is often overly conservative. Lubin et al. [6] linearized the power flow model at the system's expected operating point based on the Gaussian assumption, deriving a linear expression for output variables as a function of input variables. Wu et al. [7] used historical output data and kernel density estimation to obtain the probability distribution of input variables. Yang et al. [8], [9] relaxed the assumptions on the probability distribution of input variables and employed fuzzy sets to represent the uncertainty in renewable energy output based on distributed robust chance constraints. The above-mentioned literature on transforming chance constraints all relies on the assumption that a linear relationship exists between output and input variables, but when the uncertainty of renewable energy output is large, this can introduce significant errors.

The iterative solving framework typically separates the D-OPF model solving process from the chance constraint verification process, solving them iteratively. Based on the results of the chance constraint verification, the system's operational boundaries are dynamically adjusted and the optimal solution of the D-OPF is updated in each iteration [10]. Given that D-OPF solving techniques are already relatively mature [11], [12], the current research focus primarily lies in how to verify the chance constraints. Zou et al. [13] adopted the Monte-Carlo (MC) method to generate a large number of uncertainty scenarios, obtaining the corresponding probability distributions of output

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variables to assess whether the chance constraints were satisfied. However, the drawback of the MC method lies in its frequent need for a large number of power flow calculations during the iterative process, resulting in a significant computational burden.

To avoid the cumbersome MC sampling process, approximation methods have been widely applied. Approximation methods primarily include the unscented transformation (UT) and point estimate method (PEM), both of which focus on approximating the probabilistic characteristics of input variables using a small number of carefully selected sample points and estimating the statistical properties of output variables through corresponding output responses. Zhang et al. [14] used the UT method to model the uncertainties of load, renewable energy and energy prices, assuming that the input variables followed non-Gaussian distributions when selecting sample points. Peres et al. [15] proposed a probabilistic power flow based on the unscented transformation and demonstrated that the UT method was simple to implement. Du et al. [16] applied the PEM method in the chance constraint verification process and estimated the moments of output variables based on sample data, subsequently reconstructing the probability distribution of the output variables. However, the linear power flow model was used during the D-OPF solving process. Schmidli et al. [17], [18] solved the D-OPF based on an AC power flow model, but in the chance constraint verification process, the power flow model at the current operating point was linearly approximated and it was assumed that renewable energy output followed a Gaussian distribution. However, for power systems, which are nonlinear systems, both the UT and PEM algorithms exhibit significant approximation errors for higher-order statistical moments of output variables [19].

While UT and PEM reduce computational cost by replacing sampling-based propagation, they rely on strong assumptions such as Gaussianity or independence of input variables. In contrast, Gaussian mixture model (GMM) provides a flexible and data-driven way to model complex input distributions, enabling accurate uncertainty quantification without restrictive assumptions. Several studies have explored GMM-based methods for S-OPF or probabilistic power flow. Sun et al. [20] employed a truncated multivariate Gaussian mixture model to capture the irregular, multimodal characteristics and correlations of wind power outputs from multiple wind farms. Gao et al. [21] constructed the joint probability density function of power flow inputs and outputs based on the GMM representation of input variables and proposed a method that enables both analytical probabilistic power flow and global sensitivity analysis. Zhang et al. [22] developed a two-stage chance-constrained optimization framework based on GMM which handled non-Gaussian forecast errors through GMM-based chance constraint modeling. Wan et al. [23] utilized GMM to model voltage deviation probabilistically for risk assessment, thereby quantifying the operational uncertainty under high renewable energy penetration. These studies demonstrate the flexibility of the GMM in uncertainty modeling. However, as the power system scales up and the complexity of stochastic sources increases, the number of Gaussian components required by the GMM grows rapidly, resulting in high computational overhead in large-scale systems.

To improve the scalability of GMM-based approaches in power system applications, recent research has also focused on component reduction strategies that simplify the mixture while preserving key distributional characteristics. Singh et al. [24] pointed out that an ideal Gaussian component reduction algorithm should meet the following three criteria: (1) the mean and covariance of the fitted model should remain unchanged or exhibit minimal deviation after reduction; (2) the reduction algorithm should be simple, efficient and robust. Assa et al. [25] proposed a GMM simplification method based on the Wasserstein distance, which minimizes changes in the density shape and better captures the geometric characteristics of the distribution. Ge et al. [26] used the Mahalanobis distance as the objective function from the perspective of curve similarity theory and applied a swarm intelligence optimization algorithm to iteratively adjust and reduce Gaussian components. Jung et al. [27] and Jung et al. [28] respectively proposed Gaussian component reduction methods based on matrix Lie groups and spectral mixture kernels.

However, we must note that the above-discussed studies are subject to some important deficiencies. First, the traditional MC algorithm incurs high computational costs, making it difficult to apply in practice. Second, in the chance constraint verification phase of the iterative solving framework, the UT algorithm only accounts for the covariance of input variables and cannot consider higher-order statistical moments (such as skewness, kurtosis, etc.), while the PEM algorithm is only applicable under the assumption of independence among input variables. Therefore, to meet the application needs of stochastic optimal power flow, there is an urgent need for a method that does not rely on additional assumptions. Furthermore, this method should be capable of handling input variables with non-Gaussian distributions and complex correlations, while accurately capturing the probability distribution of output variables with high precision. Although Shargh et al. [29], [30] proposed an improved method based on Nataf transformation for spatial sampling, the high nonlinearity of the model still has a significant impact on the accuracy. Moreover, the GMM reduction algorithm needs to further balance computational efficiency and distributional fidelity.

Based on the above discussion, this paper develops an S-OPF solution framework that integrates the GMM with the UT method. The proposed framework first establishes a general two-stage structure for handling uncertainty in S-OPF. Then, GMM is adopted to represent the probabilistic characteristics of input variables, while UT is employed to propagate these uncertainties through nonlinear power flow mappings within each Gaussian component. Finally, a Gaussian component simplification model based on the Kullback-Leibler (KL) divergence is designed to accelerate the solving process. The contributions of this paper are as follows:

1) A structured combination of GMM and UT is used to approximate the output probability density function of system states, allowing accurate uncertainty quantification with significantly reduced computational cost compared to sample-based methods.

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2) A KL-divergence-guided component reduction scheme is introduced to simplify the GMM while preserving key statistical properties, improving computational scalability without compromising accuracy.

3) The proposed method is evaluated on multiple IEEE benchmark systems, demonstrating superior efficiency and accuracy compared to standard MC and various other algorithms.

Our algorithm is expected to provide system operators with a powerful tool to effectively balance economic efficiency and operational reliability in power systems with high renewable energy penetration. The rest of this paper is organized as follows: Section II introduces the construction of the S-OPF model. Section III presents the solution algorithm. Section IV provides the case studies. Section V concludes the paper.

II. Preliminaries

This section clarifies the basic S-OPF model and then elaborates on the fundamental two-stage iterative solution framework.

A. Modeling of S-OPF

In this paper, the random sources only consider the uncertainty in the output of renewable energy units. Let Ω_r denotes the set of all renewable energy units, with the number of elements in the set being *N*. The active power output of all renewable energy units in the system can be represented by the random vector $\boldsymbol{\zeta}:=[\zeta_1, \zeta_2, \ldots, \zeta_N]$, where the random variable ζ_i represents the active power output of renewable energy unit *i*. The power balance equations at the nodes of the power system are expressed as follows:

$$v_i \sum_{j\in\Omega_b} v_j \left(G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}\right) = \sum_{j\in\Omega_i^g} p_{g,j} + \sum_{j\in\Omega_i^r} \mathbb{E}(\zeta_j) - p_{d,i}, \forall i \in \Omega_b \tag{1}$$

$$v_i \sum_{j\in\Omega_b} v_j \left(G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}\right) = \sum_{j\in\Omega_i^g} q_{g,j} + \sum_{j\in\Omega_i^r} \gamma\mathbb{E}(\zeta_j) - q_{d,i}, \forall i \in \Omega_b \tag{2}$$

where Ω_i^g and Ω_i^r represent the sets of conventional generators and renewable energy units connected to node *i* respectively, Ω_b denotes the set of system nodes, $p_{d,i}$ and $q_{d,i}$ represent the active and reactive loads at node *i* respectively, G_{ij} and B_{ij} denote the real and imaginary parts of the corresponding elements in the node admittance matrix respectively, γ is the reactive/active power ratio for renewable energy units which is assumed to be constant in this paper, $\mathbb{E}(\zeta_j)$ represents the expected active output of the renewable energy unit, v_i and θ_i represent the voltage magnitude and phase angle at node *i* respectively, $\theta_{ij} = \theta_i - \theta_j$ denotes the phase angle difference between nodes *i* and *j*, $p_{g,i}$ and $q_{g,i}$ represent the active and reactive power outputs of conventional generator *i* respectively.

In (1) and (2), expected value $\mathbb{E}(\zeta_j)$ instead of the original random variable ζ_j is used to determine the baseline operating point of the system under a deterministic scenario, which corresponds to the first stage of the iterative solution framework described in Section II-B. The randomness of ζ_j is addressed in the second stage. Using the expected value ensures both the solvability of the deterministic optimal power flow problem and the effective initialization of the iterative algorithm.

To respond to the disturbances caused by the randomness of renewable energy output, the active power output of dispatchable conventional generators at the PV nodes and balance nodes needs to be adjusted to maintain power balance. Specifically, the active power output of conventional generators is adjusted through automatic generation control (AGC) as follows:

$$p_{g,i}(\boldsymbol{\zeta}) = \begin{cases} p_{g,i} + \alpha_i \sum\limits_{j\in\Omega_i^r} (\zeta_j - \mathbb{E}(\zeta_j)), i \in (\Omega_{PV} \cup \Omega_{PQ}) \\ p_{g,i} + \alpha_i \sum\limits_{j\in\Omega_i^r} (\zeta_j - \mathbb{E}(\zeta_j)) + \delta_{p,i}, i \in \Omega_{Ref} \end{cases} \tag{3}$$

where Ω_{PV} represents the set of PV nodes, Ω_{PQ} represents the set of PQ nodes, Ω_{Ref} represents the balance node, $\delta_{p,i}$ represents the additional active power loss generated after the system state change and α_i denote the AGC adjustment coefficient, which needs to satisfy $0 \le \alpha_i \le 1$ and $\sum_{i\in\Omega_g} \alpha_i = 1$.

To ensure the system's reactive power balance and to maintain the voltage magnitude at the corresponding nodes, the conventional generators at the PV nodes and balance nodes need to adjust their reactive power outputs synchronously. Whereas conventional generators at the PQ nodes keep their output constant [17], as follows:

$$q_{g,i}(\boldsymbol{\zeta}) = \begin{cases} q_{g,i} + \delta_{q,i}, i \in (\Omega_{PV} \cup \Omega_{Ref}) \\ q_{g,i}, i \in \Omega_{PQ} \end{cases} \tag{4}$$

where $\delta_{q,i}$ represents the additional reactive power loss generated after the system state change.

Additionally, according to (1)–(2), the uncertainty in the node injection power leads to uncertainty in the node voltage magnitude and phase angle, which also causes variations in the branch flow. At this point, all variables in the system can be represented as random variables. Since the random source is the active power output of the renewable energy units denoted as $\boldsymbol{\zeta}$, the random variables in the system can be denoted as $p_{g,i}(\boldsymbol{\zeta})$, $q_{g,i}(\boldsymbol{\zeta})$, $\theta_i(\boldsymbol{\zeta})$, $v_i(\boldsymbol{\zeta})$ and $i_{ij}(\boldsymbol{\zeta})$, respectively. Here, $i_{ij}(\boldsymbol{\zeta})$ represents the transmission current on the line, which can also be represented as a random variable. Based on the above analysis, the S-OPF problem can be formulated [31], as follows:

$$\min \mathrm{OBJ} = \mathbb{E}\left[\sum_{i\in\Omega_g} \left(C_i^a p_{g,i}^2(\boldsymbol{\zeta}) + C_i^b p_{g,i}(\boldsymbol{\zeta}) + C_i^c\right)\right] \tag{5}$$

$$v_i(\boldsymbol{\zeta}) \sum_{j\in\Omega_b} v_j(\boldsymbol{\zeta}) \left(G_{ij}\cos\theta_{ij}(\boldsymbol{\zeta}) + B_{ij}\sin\theta_{ij}(\boldsymbol{\zeta})\right) = \sum_{j\in\Omega_i^g} p_{g,j}(\boldsymbol{\zeta}) + \sum_{j\in\Omega_i^r} \zeta_j - p_{d,i}, \forall i \in \Omega_b \tag{6}$$

$$v_i(\boldsymbol{\zeta}) \sum_{j\in\Omega_b} v_j(\boldsymbol{\zeta}) \left(G_{ij}\sin\theta_{ij}(\boldsymbol{\zeta}) - B_{ij}\cos\theta_{ij}(\boldsymbol{\zeta})\right)$$

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$$= \sum_{j\in\Omega_i^g} q_{g,j}(\boldsymbol{\zeta}) + \sum_{j\in\Omega_i^r} \gamma\zeta_j - q_{d,i}, \forall i \in \Omega_b \tag{7}$$

$$v_i^{\min} \le v_i \le v_i^{\max}, \forall i \in (\Omega_{PV} \cup \Omega_{Ref}) \tag{8}$$

$$\mathbb{P}(p_{g,i}(\boldsymbol{\zeta}) \ge p_{g,i}^{\min}) \ge 1-\varepsilon_P, \forall i \in \Omega_b \tag{9}$$

$$\mathbb{P}(p_{g,i}(\boldsymbol{\zeta}) \le p_{g,i}^{\max}) \ge 1-\varepsilon_P, \forall i \in \Omega_b \tag{10}$$

$$\mathbb{P}(q_{g,i}(\boldsymbol{\zeta}) \ge q_{g,i}^{\min}) \ge 1-\varepsilon_Q, \forall i \in (\Omega_{PV} \cup \Omega_{Ref}) \tag{11}$$

$$\mathbb{P}(q_{g,i}(\boldsymbol{\zeta}) \le q_{g,i}^{\max}) \ge 1-\varepsilon_Q, \forall i \in (\Omega_{PV} \cup \Omega_{Ref}) \tag{12}$$

$$\mathbb{P}(v_i(\boldsymbol{\zeta}) \ge v_i^{\min}) \ge 1-\varepsilon_V, \forall i \in \Omega_{PQ} \tag{13}$$

$$\mathbb{P}(v_i(\boldsymbol{\zeta}) \le v_i^{\max}) \ge 1-\varepsilon_V, \forall i \in \Omega_{PQ} \tag{14}$$

$$\mathbb{P}(i_{ij}(\boldsymbol{\zeta}) \le i_{ij}^{\max}) \ge 1-\varepsilon_I, \forall ij \in \Omega_l \tag{15}$$

where Ω_g represent the set of conventional generators, Ω_l represents the set of transmission lines, C_i^a, C_i^b and C_i^c represent the quadratic, linear and constant coefficients of the generation cost function for conventional generator i respectively, $p_{g,i}^{\max}$, $p_{g,i}^{\min}$ and $q_{g,i}^{\max}$, $q_{g,i}^{\min}$ represent the upper and lower limits of the active and reactive power outputs of conventional generator i respectively, $v_i^{\min}$ and $v_i^{\max}$ represent the lower and upper limits of the voltage magnitude at node i respectively, $i_{ij}^{\max}$ represents the upper limit of the transmission current on line ij. ε_P, ε_Q, ε_V and ε_I represent the violation probabilities of the generator's active power output, generator's reactive power output, node voltage magnitude and branch current magnitude exceeding their limits respectively.

The objective function (5) aims to minimize the expected operational scheduling cost of conventional generators. Equation (8) imposes strict upper and lower voltage limits on the PV nodes. Equations (9)–(15) apply chance constraints to all affected variables in the system, with the required confidence level $1-\varepsilon_{P/Q/V/I}$.

To explain more concisely, define the variable set $\boldsymbol{x}$:=$\{p_{g,i}, q_{g,i}, v_j, i_{mn} | i \in (\Omega_{PV,Ref}), \forall j \in \Omega_{PQ}, \forall mn \in \Omega_l\}$, which includes all the variables in the optimal power flow model that are affected by the uncertainty in the output of renewable energy. The chance constraints (9)–(15) can be uniformly expressed as (16)–(17):

$$\mathbb{P}(x_i(\boldsymbol{\zeta}) \ge x_i^{\min}) \ge 1-\varepsilon_i \tag{16}$$

$$\mathbb{P}(x_i(\boldsymbol{\zeta}) \le x_i^{\max}) \ge 1-\varepsilon_i \tag{17}$$

B. A Two-Stage Generalized Iterative Solution Framework

Solving the S-OPF problem remains challenging due to the high coupling between the chance constraints and the AC power flow model. The literature review has already mentioned the iterative solution framework, which decouples the processing of chance constraints and the solution of the AC optimal power flow into two stages [18]. This paper continues to follow this iterative approach and provides a detailed description of the two stages in the iterative process, with the second stage being the primary focus of this paper.

Stage 1: Solving D-OPF. The core idea of this stage is to solve the D-OPF without considering the uncertainty in renewable energy output in order to obtain the system operating point for the current iteration. Considering that the interior point method (IPM) has stronger robustness and better convergence performance [12], this paper adopts the IPM algorithm as the solution method for this stage.

Stage 2: Chance constraint transformation and update. The core idea of this stage is to consider the uncertainty in renewable energy output at a given system operating point and to check whether all chance constraints are satisfied at the required confidence level. If any constraints are violated, the corresponding upper and lower bounds of the variables in the deterministic model are updated [32], [33]. Specifically, we adopt a heuristic correction strategy based on quantiles as introduced in [18], where the quantiles of the random variables and their distances to constraint bounds are evaluated in each iteration. Based on these, the feasible domain of the variables is iteratively adjusted to improve constraint satisfaction. This correction mechanism is integrated into our GMM-based uncertainty quantification framework to ensure consistency between probabilistic input modeling and constraint enforcement.

First, define the upper and lower bound tightening amounts of $\boldsymbol{x}$ in the k-th iteration as $\Delta\boldsymbol{x}_{\max}^{(k)}$ and $\Delta\boldsymbol{x}_{\min}^{(k)}$. The calculation methods for each variable are as follows:

$$\Delta x_{i,\max}^{(k)} = \left[Q_{1-\varepsilon_i}^{(k)}(x_i) - x_{i,\max}\right]^+ \tag{18}$$

$$\Delta x_{i,\min}^{(k)} = \left[x_{i,\min} - Q_{1-\varepsilon_i}^{(k)}(x_i)\right]^+ \tag{19}$$

where $Q_\alpha^{(k)}(x_i)$ denotes the α-th quantile of variable $x_i(\boldsymbol{\zeta})$ in the k-th iteration and $[\cdot]^+ = \max\{\cdot, 0\}$. Therefore, in the ($k$+1)-th iteration, the upper and lower bounds of variable $\boldsymbol{x}$ are tightened as follows:

$$\boldsymbol{x}_{\max}^{(k+1)} = \boldsymbol{x}_{\max}^{(k)} - \Delta\boldsymbol{x}_{\max}^{(k)} \tag{20}$$

$$\boldsymbol{x}_{\min}^{(k+1)} = \boldsymbol{x}_{\min}^{(k)} + \Delta\boldsymbol{x}_{\min}^{(k)} \tag{21}$$

The core of this correction strategy lies in calculating $Q_\alpha^{(k)}(x_i)$ and the key to solving $Q_\alpha^{(k)}(x_i)$ is accurately characterizing the probability distribution of $\boldsymbol{x}(\boldsymbol{\zeta})$, which is elaborated upon in the following sections as the focus of this paper. It is worth noting that the probability distribution of $\boldsymbol{x}(\boldsymbol{\zeta})$ relies on historical samples of renewable energy output $\{\zeta^{(i)}\}$, avoiding any prior assumptions about the probabilistic characteristics of renewable energy output. At the same time, it fully considers the correlation between the outputs of different renewable energy units, making the method proposed in this paper highly universal.

III. Solution Algorithm

Section II has transformed the solution of the S-OPF into the characterization of the probability distribution of $\boldsymbol{x}(\boldsymbol{\zeta})$. The task of this section is to handle the complex input variables $\boldsymbol{\zeta}$ and obtain the probability distribution of the output variables $\boldsymbol{x}(\boldsymbol{\zeta})$ with higher precision. To this end, we combine GMM and UT in a two-step strategy. First, GMM is used to flexibly model the non-Gaussian input uncertainties. Then, UT is applied to each Gaussian component to propagate uncertainty through the nonlinear power flow model. This enables accurate and efficient

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reconstruction of the full output probability density function while preserving system nonlinearity.

A. Input Variable Modeling Based on GMM

In power systems, random variables often exhibit non-Gaussian characteristics and strong correlations. Using a single standard probability distribution function for fitting can lead to significant errors, which in turn affects the accuracy of the probability modeling of output variables [34]. Given that GMM excels at smoothly approximating arbitrarily shaped density functions—particularly for modeling complex, multi-modal and correlated random variables [21], [35], this chapter proposes using a GMM to model the multivariate random variables $\boldsymbol{\zeta}:=[\zeta_1, \zeta_2, \ldots, \zeta_N]$.

For an N-dimensional random vector $\boldsymbol{\zeta}$, its joint probability density function $f(\boldsymbol{\zeta})$ can be approximated by a linear combination of several (denoted as N_G) N-dimensional Gaussian distributions:

$$f(\boldsymbol{\zeta}) = \sum_{i=1}^{N_G} \omega_i \mathcal{N}(\boldsymbol{\zeta}; \boldsymbol{\mu}_i, \boldsymbol{P}_i) \tag{22}$$

where ω_i, $\boldsymbol{\mu}_i$ and $\boldsymbol{P}_i$ represent the weight, mean and covariance of the i-th Gaussian component respectively. $\mathcal{N}(\boldsymbol{\zeta}; \boldsymbol{\mu}_i, \boldsymbol{P}_i)$ denotes the Gaussian distribution of the ith component of $\boldsymbol{\zeta}$ and its probability density function is given by:

$$f_{\mathcal{N}}(\boldsymbol{\zeta}; \boldsymbol{\mu}_i, \boldsymbol{P}_i) = \frac{1}{\sqrt{(2\pi)^N \det \boldsymbol{P}_i}} \cdot \exp\left(-\frac{1}{2}(\boldsymbol{\zeta} - \boldsymbol{\mu}_i)^\top \boldsymbol{P}_i^{-1}(\boldsymbol{\zeta} - \boldsymbol{\mu}_i)\right), i = 1, 2, \cdots N_G \tag{23}$$

where $\det$ calculates the determinant of the matrix.

The weights of each Gaussian component must satisfy the normalization condition, that is:

$$0 \le \omega_i \le 1, \sum_{i=1}^{N_G} \omega_i = 1 \tag{24}$$

In (22), the parameters to be determined for the GMM include the number of components N_G and the parameters ω_i, $\boldsymbol{\mu}_i$ and $\boldsymbol{P}_i$ for each component. If N_G is fixed, the remaining three parameters can be estimated based on historical sample data using the expectation maximization (EM) algorithm [24].

Based on the above discussion, the key challenge of GMM modeling is how to determine N_G. Taking Case B in Section IV as an example, we perform GMM fitting with the number of Gaussian components set to $N_G = 3, 8, 20$, respectively. Fig. 1 shows the joint probability density of the output from photovoltaic plant (PVP) 1 and PVP 2 under different numbers of Gaussian components obtained through GMM fitting. By comparing Fig. 1(a), (c) and (d), it can be observed that the GMM is able to successfully capture the non-Gaussian shape, strong correlations and multi-modal characteristics present in the input data distribution. This further supports the rationale for using GMM to fit random variables with arbitrary distributions. Clearly, increasing the number of Gaussian components can better reflect the statistical characteristics of the input variables, thereby improving the fitting accuracy. However, this also increases the computational complexity [36]. Therefore, to balance the trade-off between model complexity and fitting accuracy, this paper uses the Akaike information criterion (AIC) to determine the number of Gaussian components. The AIC performance metric L_{AIC} is defined as follows:

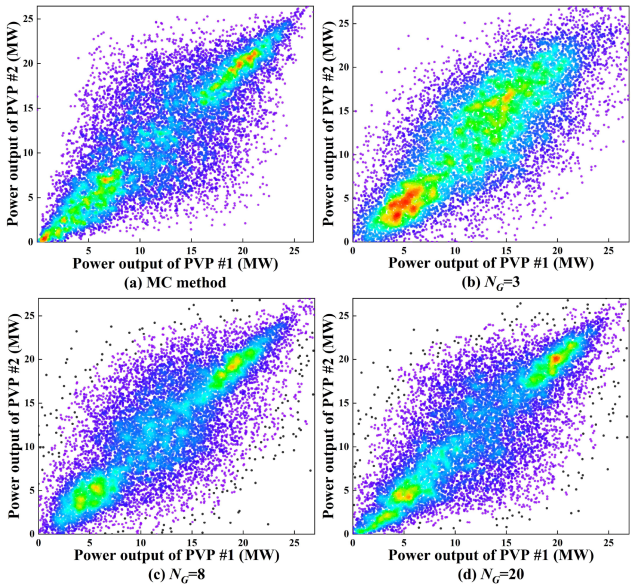


Fig. 1. Joint probability density plot of active power output between PVP 1 and PVP 2.

$$L_{\mathrm{AIC}} = 2N_G - 2\ln(\arg\max L_f) \tag{25}$$

where $\arg\max L_f$ represents the likelihood function. In general, as the number of model components N_G increases, L_{AIC} continuously decreases. When the number of components becomes large enough, the rate of decline in L_{AIC} gradually slows down. Therefore, when selecting the number of model components, the value of N_G should be chosen such that L_{AIC} is sufficiently small and no longer significantly decreases. In summary, the procedure for determining the number of GMM components can be summarized as follows:

1) Set the initial value of the number of Gaussian components N_G;
2) Estimate the parameters and corresponding weights of each Gaussian component using the EM algorithm as described in [24];
3) Calculate the AIC value corresponding to the current GMM model according to (25) and denote it as L_{AIC,N_G};
4) Set $N_G \leftarrow N_G + 1$ and return to step (2);
5) Repeat the above steps until $|L_{\mathrm{AIC},N_G+1} - L_{\mathrm{AIC},N_G}| < \delta_{\mathrm{AIC}}$ and $L_{\mathrm{AIC},N_G+1} < \tau_{\mathrm{AIC}}$, δ_{AIC} and τ_{AIC} are the given thresholds.

B. Computation of Output Variable Statistics Based on the UT Algorithm

Section III-B focuses on how to calculate the statistical characteristics of the output variables $\boldsymbol{x}(\boldsymbol{\zeta})$ based on the statistical features of the input variables $\boldsymbol{\zeta}$ i.e., the random sources. The implicit mathematical function of $\boldsymbol{x}$ can be expressed as follows:

$$\boldsymbol{x} = h\left(\boldsymbol{y}^{\mathrm{fixed}}, \boldsymbol{\zeta}\right) \tag{26}$$

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where $\boldsymbol{y}^{\text{fixed}}$ represents the system operating point obtained in each iteration of *Stage 1* in Section II-B. The mapping h corresponds to the AC power flow equations, which are nonlinear and implicit. As a result, computing the output distribution $f(\boldsymbol{x})$ analytically is infeasible.

Let the mean of the random vector $\boldsymbol{\zeta}$ be $\boldsymbol{\mu}_\xi$ and the covariance matrix be $\boldsymbol{P}_\xi$. The traditional method for characterizing statistical features is the UT algorithm [37], which selects a small number of structured sigma points based on fixed rules to ensure that the first two-order statistical moments of the samples match those of the input variables. The UT procedure consists of the following three steps:

Step 1: Determine the positions of the selected sample points according to the following formula:

$$\zeta_{[i]} = \begin{cases} \boldsymbol{\mu}_\xi, i = 0 \\ \boldsymbol{\mu}_\xi + \sqrt{N+\lambda}\left(\sqrt{\boldsymbol{P}_\xi}\right)_{[i]}, i = 1, 2, \ldots, N \\ \boldsymbol{\mu}_\xi - \sqrt{N+\lambda}\left(\sqrt{\boldsymbol{P}_\xi}\right)_{[i-N]}, i = N+1, \ldots, 2N \end{cases} \tag{27}$$

where $\sqrt{\boldsymbol{P}_\xi}$ is the square root of the matrix $\boldsymbol{P}_\xi$, $\left(\sqrt{\boldsymbol{P}_\xi}\right)_{[i]}$ represents the i-th column vector of the matrix $\boldsymbol{P}_\xi$, λ is a tunable parameter, typically set as $\lambda = 3 - N$. *Step 1* generates $2N+1$ sample points based on the given mean and covariance.

The mean weights $\omega_{[i]}$ and covariance weights $\tilde{\omega}_{[i]}$ corresponding to each sample point are calculated as follows:

$$\omega_{[i]} = \tilde{\omega}_{[i]} = \begin{cases} \frac{\lambda}{N+\lambda}, i = 0 \\ \frac{1}{2(N+\lambda)}, i = 1, 2 \ldots, 2N \end{cases} \tag{28}$$

The UT weights $\omega_{[i]}$ and $\tilde{\omega}_{[i]}$ are identical across all components because they are part of the standard UT method [37], designed to precisely match the first two moments of the input distribution.

Step 2: Input each sample point into the nonlinear mapping function and calculate the corresponding output samples:

$$\boldsymbol{x}_{[i]} = h\left(\boldsymbol{y}^{\text{fixed}}, \boldsymbol{\zeta}_{[i]}\right) \tag{29}$$

Step 3: Calculate the mean and covariance matrix of the output variables:

$$\boldsymbol{\mu}_{\boldsymbol{x}} = \sum_{i=0}^{2N} \omega_{[i]} \boldsymbol{x}_{[i]} \tag{30}$$

$$\boldsymbol{P}_{\boldsymbol{x}} = \sum_{i=0}^{2N} \tilde{\omega}_{[i]} \left(\boldsymbol{x}_{[i]} - \boldsymbol{\mu}_x\right)\left(\boldsymbol{x}_{[i]} - \boldsymbol{\mu}_x\right)^{\mathrm{T}} \tag{31}$$

Based on the analysis above, the UT algorithm is easy to implement as it requires only $2N+1$ power flow calculations to obtain the first two statistical moments of the output variables [38]. However, for *Stage 2* in Section III-B, relying solely on the UT algorithm two major drawbacks:

1) When the input variables exhibit significant non-Gaussian characteristics, the accuracy of the UT algorithm in estimating the statistical moments of the output variables may decrease.

2) When the input variables exhibit significant non-Gaussian characteristics, it is difficult to fully capture the probabilistic features of the output variables using only the mean and variance. Taking Case A in Section IV as an example, Fig. 2 illustrates

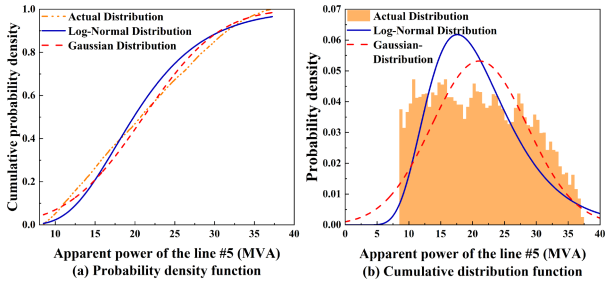


Fig. 2. Comparison of common bivariate distributions and actual distributions.

the probability distribution of the apparent power on line 5 in the IEEE 14-bus system. When only the output mean and variance are given, common two-parameter distributions such as the Gaussian and log-normal distributions result in significant fitting errors compared to the actual probability distribution. This indicates that relying solely on the UT algorithm is insufficient to meet the requirements of *Stage 2*.

C. GMM-UT-Based Output Variable Probability Density Reconstruction

As mentioned in Section III-B, directly applying standard two-parameter distributions to fit the output variables often leads to significant errors. An intuitive solution is to introduce the GMM model to fit the probability distribution of the output variables, similar to the approach used for modeling the input variables. Therefore, the goal of Section III-C is to integrate the GMM model with the UT algorithm, enabling efficient and accurate approximation of $f(\boldsymbol{x})$ under high-dimensional and non-Gaussian input uncertainty.

In fact, (22) and (24) can be interpreted as the existence of N_G Gaussian components, where the probability of the i-th Gaussian component occurring is ω_i. Therefore, let us denote by Φ_i the probabilistic event that the input $\boldsymbol{\zeta}$ is drawn from the i-th Gaussian component in the mixture model. The corresponding mean and covariance of this component are $\boldsymbol{\mu}_i$ and $\boldsymbol{P}_i$, respectively, and the probability of event Φ_i occurring is given by $\mathbb{P}(\Phi_i)$. Based on the law of total probability, (22) can then be equivalently rewritten as:

$$f(\boldsymbol{\zeta}) = \sum_{i=1}^{N_G} \mathbb{P}(\Phi_i) f(\boldsymbol{\zeta}|\Phi_i) \tag{32}$$

$$\begin{cases} \boldsymbol{\zeta}\,|\Phi_i \sim f(\boldsymbol{\zeta}\,|\Phi_i) = \mathcal{N}(\boldsymbol{\zeta}; \boldsymbol{\mu}_i, \boldsymbol{P}_i) \\ \mathbb{P}(\Phi_i) = \omega_i \end{cases} \tag{33}$$

where $f(\boldsymbol{\zeta}|\Phi_i)$ is the conditional probability distribution under the event Φ_i.

In (32), Φ_i is introduced to highlight the mixture nature of the GMM, where each component represents a latent scenario under which the input variable $\boldsymbol{\zeta}$ follows a specific Gaussian distribution. Since the overall distribution of $\boldsymbol{\zeta}$ is expressed as a weighted sum of these Gaussian components, the number of events equals the number of Gaussian components N_G in the GMM used to model the uncertainties of renewable energy.

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Algorithm 1: Probability Modeling of Output Variables.

Input: GMM Model of $\boldsymbol{\zeta}$: $f(\boldsymbol{\zeta}) = \sum_{i=1}^{N_G} \omega_i \mathcal{N}(\boldsymbol{\zeta}; \boldsymbol{\mu}_i, \boldsymbol{P}_i)$

1: **For** $l = 1, 2, \ldots, N_G$ **do**

2: Calculate the sampling point location $\boldsymbol{\xi}_{[i]}^{(l)}$ and weights $\omega_{[i]}^{(l)}$, $\tilde{\omega}_{[i]}^{(l)}$

3: Call (28) to calculate the output variable samples $\boldsymbol{x}_{[i]}^{(l)}$

4: Calculate the mean and covariance of the output components: $\boldsymbol{\mu}_\chi^{(l)} = \sum_{i=0}^{2N} \omega_{[i]}^{(l)} \boldsymbol{x}_{[i]}^{(l)} \boldsymbol{P}_\chi^{(l)} = \sum_{i=0}^{2N} \tilde{\omega}_{[i]}^{(l)} (\boldsymbol{x}_{[i]}^{(l)} - \boldsymbol{\mu}_x^{(l)})(\boldsymbol{x}_{[i]}^{(l)} - \boldsymbol{\mu}_x^{(l)})^{\mathrm{T}}$

5: **End for**

Output: Probability distribution of $\boldsymbol{x}$: $f(\boldsymbol{x}) = \sum_{i=1}^{N_G} \omega_i \mathcal{N}(\boldsymbol{x}; \boldsymbol{\mu}_\chi^i, \boldsymbol{P}_\chi^i)$

Starting from (32), if the conditional probability distribution $f(\boldsymbol{\zeta}|\Phi_i)$ of the output variable $\boldsymbol{x}$ under event Φ_i is known, the complete probability distribution $f(\boldsymbol{x})$ of $\boldsymbol{x}$ can be obtained using the law of total probability. Here, it is assumed that for each Gaussian component in the GMM model of the input variables, the corresponding output component, after a nonlinear transformation, can also be approximated by a Gaussian distribution.

Remark 1: The previously mentioned "Gaussian-in, Gaussian-out" approximation has become a common principle in benchmark algorithms across many mature fields. A typical example is the unscented Kalman filter, which has been successfully applied for decades in safety-critical systems. Its stable and reliable performance provides strong external validation for the soundness of this approximation [39], [40].

Based on this assumption, the probability distribution $f(\boldsymbol{x})$ of the output variable $\boldsymbol{x}$ can be expressed as follows using the law of total probability:

$$f(\boldsymbol{x}) = \sum_{i=1}^{N_G} \omega_i \mathcal{N}\left(\boldsymbol{x}; \boldsymbol{\mu}_\chi^i, \boldsymbol{P}_\chi^i\right) \tag{34}$$

where $\boldsymbol{\mu}_\chi^i$ and $\boldsymbol{P}_\chi^i$ represent the mean and covariance of the output variable when event Φ_i occurs, respectively. At this point, the key challenge is to calculate $\boldsymbol{\mu}_\chi^i$ and $\boldsymbol{P}_\chi^i$. This can clearly be addressed using the UT algorithm from Section III-B, which generates the GMM model for the output variable. That is to say, while the overall distribution of $\boldsymbol{\zeta}$ can be non-Gaussian, each component of $\boldsymbol{\zeta}$ is locally Gaussian, which enables the use of UT for uncertainty propagation.

The process described above can be defined as Algorithm 1. Algorithm 1 adopts a sample point selection strategy based on GMM-UT, which requires only $(2N+1) \times N_G$ power flow calculations to construct the complete probability distribution of the output variables, significantly improving computational efficiency compared to MC methods. Algorithm 1 can theoretically approximate the input and output variables with arbitrary distributions using the GMM model without requiring any assumptions about their probability distributions.

Note: Steps 1 to 5 need to be executed for each Gaussian component $\boldsymbol{\zeta}|\Phi_i$.

D. Accelerated Solution Strategy Based on GMM Simplification

As the dimensions N of the random variable $\boldsymbol{\zeta}$ increases, the number of Gaussian components N_G required to ensure the accuracy of the GMM fitting also increases, leading to a rapid rise in the computational complexity of Algorithm 1. To alleviate the computational burden of Algorithm 1, Section III-D explores how to reduce the number of Gaussian components while ensuring the fitting accuracy. Unlike principal component analysis or singular value decomposition, which reduce the number of variables via projection, our component reduction method keeps the variable dimension fixed and focuses on minimizing the number of Gaussian components to reduce computation while preserving the multi-modal distribution structure.

This paper designs a multi-round pairing and merging method. For ease of reference in the following sections, let $(\omega_i, \boldsymbol{\mu}_i, \boldsymbol{P}_i)$ represent a Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{P}_i)$ with weights ω_i. In order to ensure that the statistical characteristics of the GMM model are not significantly affected after reducing the number of components, the new components formed by merging should match the zero-order moment, first-order moment and second-order moment of the original two components, according to the moment matching principle. That is, for two components $\{(\omega_i, \boldsymbol{\mu}_i, \boldsymbol{P}_i), (\omega_j, \boldsymbol{\mu}_j, \boldsymbol{P}_j)\}$, the newly merged component $(\omega_{ij}, \boldsymbol{\mu}_{ij}, \boldsymbol{P}_{ij})$ should satisfy the following conditions:

$$\omega_{ij} = \omega_i + \omega_j \tag{35}$$

$$\boldsymbol{\mu}_{ij} = \omega_{i|ij}\boldsymbol{\mu}_i + \omega_{j|ij}\boldsymbol{\mu}_j \tag{36}$$

$$\boldsymbol{P}_{ij} = \omega_{i|ij}\omega_{j|ij}\left(\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\right)\left(\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\right)^{\mathrm{T}} + \omega_{i|ij}\boldsymbol{P}_i + \omega_{j|ij}\boldsymbol{P}_j \tag{37}$$

where $\omega_{i|ij} = \frac{\omega_i}{\omega_i + \omega_j}$, $\omega_{j|ij} = \frac{\omega_j}{\omega_i + \omega_j}$

If we need to reduce a GMM consisting of N_G components to one with M_G components ($M_G < N_G$), then a total of $(N_G - M_G)$ pairings are required. To minimize the fitting error introduced by the component reduction, a criterion must be designed to select the best two components for merging, so that the difference between the resulting simplified model $f^{M_G}(\boldsymbol{\zeta})$ and the original model $f^{N_G}(\boldsymbol{\zeta})$ is minimized, where the superscript denotes the number of Gaussian components. This paper chooses the KL divergence-based criterion to select the components i and j that need to be paired in each round of merging. The KL divergence between the probability distributions $f_i(\boldsymbol{\zeta})$ and $f_j(\boldsymbol{\zeta})$ is defined as:

$$d_{\mathrm{KL}}(f_i(\boldsymbol{\zeta}), f_j(\boldsymbol{\zeta})) = \int_{\mathbb{R}^N} f_i(\boldsymbol{\zeta}) \log \frac{f_i(\boldsymbol{\zeta})}{f_j(\boldsymbol{\zeta})} d\boldsymbol{\zeta} \tag{38}$$

If the effect of component pairing and merging is directly evaluated using (38), it results in a significant computational burden. Reference [41] proves the following two theorems which are used to derive the upper bound of the KL divergence between the two Gaussian mixture distributions before and after merging to address this issue.

Theorem 1: If there exist arbitrary probability distributions $g_1(\boldsymbol{\zeta})$, $g_2(\boldsymbol{\zeta})$ and $g_3(\boldsymbol{\zeta})$ all with N-dimensional space, and given that $0 \le \rho \le 1$ and $\tilde{\rho} = 1 - \rho$, then: $d_{\mathrm{KL}}(\rho g_1 + \tilde{\rho} g_3, \rho g_2 + \tilde{\rho} g_3) \le \rho d_{\mathrm{KL}}(g_1, g_2)$.

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Theorem 2: Under the same conditions as Theorem 1, we obtain $d_{\mathrm{KL}}(\rho g_1+\tilde{\rho} g_2, g_3) \leq \rho d_{\mathrm{KL}}(g_1, g_3)+\tilde{\rho} d_{\mathrm{KL}}(g_2, g_3)$ and $d_{\mathrm{KL}}(g_3, \rho g_1+\tilde{\rho} g_2) \leq \rho d_{\mathrm{KL}}(g_3, g_1)+\tilde{\rho} d_{\mathrm{KL}}(g_3, g_2)$.

For a GMM model composed of k+1 components, let $f_{k|ij}^{k+1}(\boldsymbol{\zeta})$ denote the GMM model composed of k components obtained by merging the i-th and j-th components from $f^{k+1}(\boldsymbol{\zeta})$. Let $g_1=\{(\omega_{i|ij}, \boldsymbol{\mu}_i, \boldsymbol{P}_i), (\omega_{j|ij}, \boldsymbol{\mu}_j, \boldsymbol{P}_j)\}$ $g_2=(1, \boldsymbol{\mu}_{ij}, \boldsymbol{P}_{ij})$ and $\rho=\omega_i+\omega_j$ represent the Gaussian mixture distributions of two components, then according to Theorem 1, the following equation holds:

$$(\omega_i+\omega_j)\, d_{\mathrm{KL}}\left(\left(\omega_{i|ij}, \boldsymbol{\mu}_i, \boldsymbol{P}_i\right), \left(\omega_{j|ij}, \boldsymbol{\mu}_j, \boldsymbol{P}_j\right), (1, \boldsymbol{\mu}_{ij}, \boldsymbol{P}_{ij})\right) \geq d_{\mathrm{KL}}(f^{k+1}, f_{k|ij}^{k+1}) \tag{39}$$

Furthermore, let $\rho=\omega_{i|ij}$, $g_1=(1, \boldsymbol{\mu}_i, \boldsymbol{P}_i)$, $g_2=(1, \boldsymbol{\mu}_j, \boldsymbol{P}_j)$ and $g_3=(1, \boldsymbol{\mu}_{ij}, \boldsymbol{P}_{ij})$, then according to Theorem 2 and combining with (39), we can infer:

$$\begin{aligned} d_{\mathrm{KL}}(f^{k+1}, f_{k|ij}^{k+1}) \leq\ & \omega_i d_{\mathrm{KL}}\left(\left(1, \boldsymbol{\mu}_i, \boldsymbol{P}_i\right), (1, \boldsymbol{\mu}_{ij}, \boldsymbol{P}_{ij})\right) \\ & + \omega_j d_{\mathrm{KL}}\left(\left(1, \boldsymbol{\mu}_j, \boldsymbol{P}_j\right), (1, \boldsymbol{\mu}_{ij}, \boldsymbol{P}_{ij})\right) \end{aligned} \tag{40}$$

We denote the right-hand side of (40) as $D(i, j)$. For any two Gaussian distributions $(1, \boldsymbol{\mu}_i, \boldsymbol{P}_i)$ and $(1, \boldsymbol{\mu}_j, \boldsymbol{P}_j)$, their KL divergence can be explicitly calculated as:

$$\begin{aligned} 2 d_{\mathrm{KL}}\left(\left(1, \boldsymbol{\mu}_i, \boldsymbol{P}_i\right), \left(1, \boldsymbol{\mu}_j, \boldsymbol{P}_j\right)\right) =\ & \log \frac{\det \boldsymbol{P}_j}{\det \boldsymbol{P}_i} \\ & + \operatorname{tr}\left(\boldsymbol{P}_j^{-1}\left[\boldsymbol{P}_i-\boldsymbol{P}_j+\left(\boldsymbol{\mu}_i-\boldsymbol{\mu}_j\right)\left(\boldsymbol{\mu}_i-\boldsymbol{\mu}_j\right)^{\mathrm{T}}\right]\right) \end{aligned} \tag{41}$$

where $\operatorname{tr}(\cdot)$ denotes the trace of the matrix.

Substituting (41) into $D(i, j)$, we obtain:

$$\begin{aligned} 2 D(i, j) =\ & \operatorname{tr}\left(\boldsymbol{P}_{ij}^{-1} \breve{\boldsymbol{P}}_{ij}\right)+\left(\omega_i+\omega_j\right) \log \det \boldsymbol{P}_{ij} \\ & -\omega_i \log \det \boldsymbol{P}_i-\omega_j \log \det \boldsymbol{P}_j \end{aligned} \tag{42}$$

$$\begin{aligned} \breve{\boldsymbol{P}}_{ij} =\ & \omega_i \boldsymbol{P}_i+\omega_j \boldsymbol{P}_j+\frac{\omega_i \omega_j}{\omega_i+\omega_j}\left(\boldsymbol{\mu}_i-\boldsymbol{\mu}_j\right)\left(\boldsymbol{\mu}_i-\boldsymbol{\mu}_j\right)^{\mathrm{T}} \\ & -\left(\omega_i+\omega_j\right) \boldsymbol{P}_{ij} \end{aligned} \tag{43}$$

Furthermore, by substituting (37) into (43), we obtain:

$$D(i, j)=\frac{1}{2}\left[\begin{array}{l} \left(\omega_i+\omega_j\right) \log \det \boldsymbol{P}_{ij}-\omega_i \log \det \boldsymbol{P}_i \\ -\omega_j \log \det \boldsymbol{P}_j \end{array}\right] \tag{44}$$

At this point, an accelerated solving strategy based on GMM simplification can be constructed, as shown in Algorithm 2. In each round of pairwise merging, calculate the corresponding $D(i, j)$ for all possible pairing combinations (i, j) and select the combination (i, j) that minimizes $D(i, j)$ for merging. This process is repeated until the number of components is reduced to the specified value M_G. Although the initial number of Gaussian components N_G is selected based on the AIC criterion to ensure sufficient statistical representation of the input distribution, this choice does not take into account the computational burden of subsequent uncertainty propagation. Therefore, Section III-D introduces a component reduction procedure that simplifies the GMM to a form with fewer components ($M_G < N_G$), thereby improving the computational efficiency of *Stage 2*.

Algorithm 2: Accelerated Solution Strategy.

Input: Model $f(\boldsymbol{\zeta})=\sum_{i=1}^{N_G} \omega_i \mathcal{N}(\boldsymbol{\zeta}; \boldsymbol{\mu}_i, \boldsymbol{P}_i)$ to be simplified and M_G

Initialize the number of components $k \leftarrow n$
While $k > m$ **do**
  Initialize the tuple $(p, q) \leftarrow (1, 2)$, $D(p, q) \leftarrow D(1, 2)$
  For $i=1, 2, \ldots, k$ **do**
    For $j=i+1, \ldots, k$ **do**
      Calculate $D(i, j)$ based on (44)
      If $D(i, j) \leq D(p, q)$ **then**
        Update the tuple $(p, q) \leftarrow (i, j)$, $D(p, q) \leftarrow D(i, j)$
      End if
    End for
  End for
  Call (35-37) to merge the p-th and q-th Gaussian components
  Update the number of components $k \leftarrow k+1$
End while

Output: The simplified GMM model $f(\boldsymbol{\zeta})=\sum_{i=1}^{M_G} \omega_i \mathcal{N}(\boldsymbol{\zeta}; \boldsymbol{\mu}_i, \boldsymbol{P}_i)$

M_G is a user-defined parameter that balances computational efficiency and modeling accuracy. A smaller M_G significantly reduces the number of AC power flow evaluations, thereby improving overall computational efficiency. However, excessive simplification may impair the GMM's ability to capture the statistical characteristics of the input variable $\boldsymbol{\zeta}$. In this text, the value of M_G is determined based on either a target component reduction ratio or a predefined threshold on cumulative KL divergence. In the latter case, we continuously monitor the KL divergence between the original and simplified models during the reduction process. If the divergence exceeds the specified threshold, the merging is terminated.

The process described above can be defined as Algorithm 2. Based on Algorithm 2, the number of power flow calculations can be reduced by $(N_G-M_G) \times (2N+1)$ times than that required in Algorithm 1, significantly improving the computational efficiency.

It is worth emphasizing that the GMM model obtained through the aforementioned simplification algorithm exhibits higher fitting accuracy compared to the GMM model with M_G components obtained directly through fitting. This phenomenon is demonstrated and analyzed in Section IV.

While the GMM simplification method proposed in this section does not guarantee an optimal reduced mixture in the theoretical sense, it enables effective simplification of high-dimensional mixtures with negligible impact on key distributional properties.

E. S-VSC-OPF Solution Algorithm

Section III-A to III-D addressed how to efficiently construct the probability density of the output variables. Next, by embedding Algorithm 2 into the general two-stage iterative

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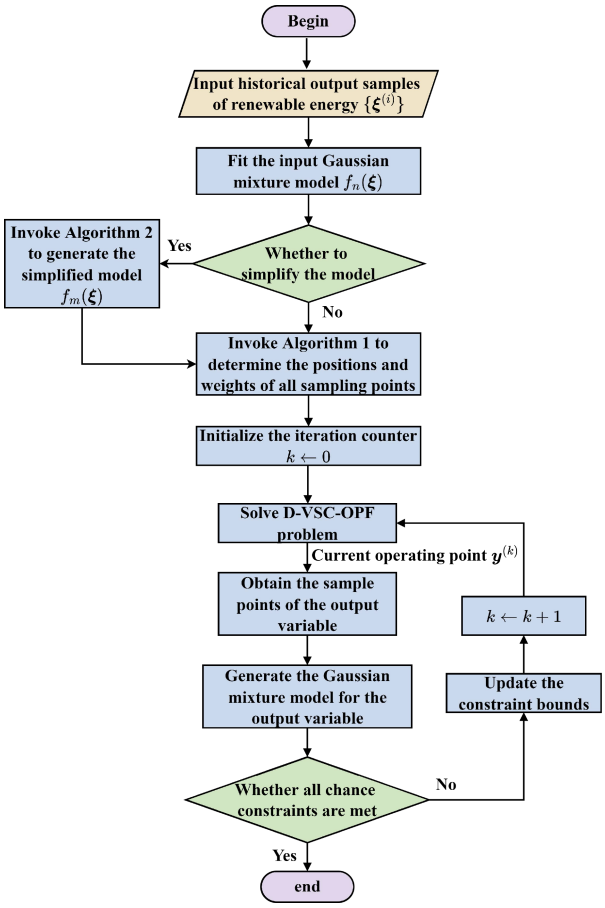


Fig. 3. Flowchart of the iterative solution algorithm for S-OPF based on GMM-UT.

solution framework outlined in Section II-B, an efficient solution algorithm for S-OPF can be developed. The algorithm flow is illustrated in Fig. 3.

IV. Case Study

This section demonstrates and analyzes the accuracy, reliability and computational efficiency of the proposed algorithm based on four electrical systems of different scales. Case A is the IEEE-14 bus system, Case B is the IEEE-30 bus system, Case C is the European 89-bus transmission system and Case D is the Illinois 200-bus system, with the last two being actual power systems. In these four cases, 3, 5, 7 and 10 PVPs are integrated as random sources in the system and the hourly historical output data of the PVPs are obtained from the platform Renewables.ninja [42]. While only photovoltaic output uncertainty is considered in the case study for clarity, the proposed framework is directly applicable to multivariate uncertainty sources, including wind power and load demand variation. The simulation environment consists of an Intel(R) Core(TM) i9-13900HX 2.20 GHz processor, 32 GB of RAM and the programming environment is MATLAB 2022a.

For the above four cases, the input random variables are fitted using a GMM based on the steps outlined in Section III-A. The performance indicator AIC, denoted as L_{AIC}, is shown as a function of the number of components in Fig. 4. According to Fig. 4, the number of components for the pre-reduced GMM model in this section is set to 10, 16, 30 and 50 for the four cases, respectively. Accordingly, for each case, the number of simulation samples used by the MC method is 8000, 12800, 24000 and 40000, respectively. The photovoltaic penetration rates for each case are 24.74%, 27.04%, 29.84% and 31.82%, respectively.

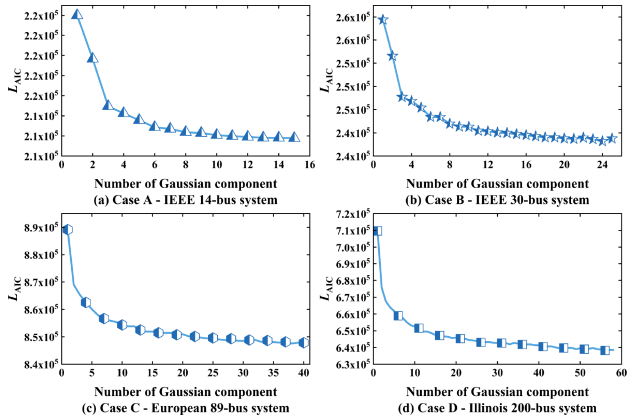


Fig. 4. Trend of L_{AIC} with respect to the number of Gaussian components.

A. Analysis of Output Probability Modeling Accuracy

Section IV-A uses the IEEE-30 bus system as an example to demonstrate the superiority of the output variable probability modeling method of the proposed GMM-UT algorithm. Specifically, in terms of accuracy, Section IV-A uses the MC-based method described in [13] as the benchmark and compares the results obtained by the following four approximation methods:

Method 1: The GMM-UT algorithm proposed in this paper.

Method 2: A probability modeling method combining the Nataf transformation, the PEM method and series expansion, as detailed in [32].

Method 3: The UT algorithm is used to obtain the first two moments of the output variables, which are then fitted to a Gaussian distribution.

Method 4: Based on the Nataf transformation, the original variables are projected into a set of independent Gaussian variables and multivariate orthogonal polynomial bases are constructed using Hermite polynomials. This method is referred to as Nataf-PCE, as detailed in [18].

Fig. 5 shows the scatter plots of sample points selected for PVP 1 - PVP 2 and PVP 3 - PVP 4 using the four methods mentioned above. Clearly, the GMM-UT algorithm proposed in this chapter, by decomposing the input variables into multiple components, covers both the regions with higher and lower joint probability densities. This allows the selected sample points to more comprehensively represent the uncertainty of the input variables, leading to a more accurate characterization of the output variables. Although the GMM-UT algorithm requires more sample points compared to the other three methods, it still significantly reduces the sample size required compared to

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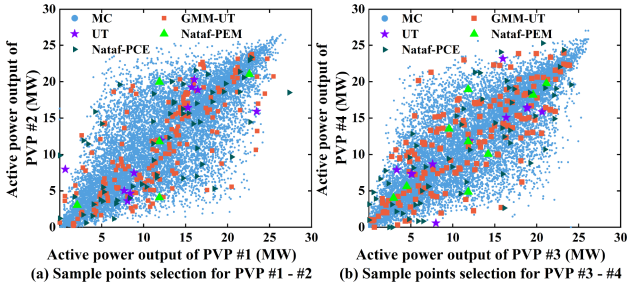


Fig. 5. Selection of sample points based on MC, GMM-UT, Nataf-PEM and UT algorithms.

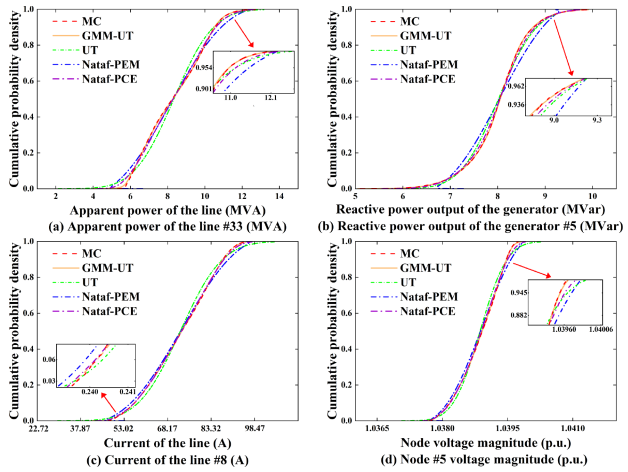


Fig. 6. Partial output variables' cumulative probability density function.

the traditional MC method, striking a good balance between accuracy and computational efficiency.

To illustrate this point, with the system operating point fixed and using the results obtained by the MC method as the benchmark, Fig. 6 presents the cumulative probability density functions of some output variables based on the results from each method. As shown in the Fig. 6, the probability distribution obtained using the GMM-UT algorithm is very close to the actual result, while the results of the other two methods show some deviation from the MC method. Therefore, the algorithm proposed in this chapter is more suitable for scenarios where the input variables follow a non-Gaussian distribution and exhibit complex correlations, compared to existing approximation methods.

For more output variables, Table I provides a further quantitative analysis of the fitting errors. The evaluation metric used is the root mean squared percentage error (RMSPE) and its calculation expression is as follows:

$$\eta = \sqrt{\frac{1}{N_{tot}} \sum\nolimits_{i=1}^{N_{tot}} \left(\frac{C_i - C_{M.i}}{C_{M.i}} \right)^2} \times 100\% \qquad (45)$$

where N_{tot} represents the total number of statistical points on the cumulative probability density function of the output variable; C_i and $C_{M,i}$ represent the values of the cumulative probability density function at the i-th point obtained from the approximation method and the MC method, respectively.

TABLE I
FITTING ERRORS OF PARTIAL OUTPUT VARIABLES UNDER DIFFERENT METHODS (RMSPE WITH 95% CONFIDENCE INTERVALS, UNIT: %)

Output variables	GMM-UT	Nata-PEM
σ	0.02 (±0.008)	0.11 (±0.028)
v_3	0.0011 (±0.0002)	0.011 (±0.002)
v_4	0.0013 (±0.0005)	0.013 (±0.002)
i_{33}	0.44 (±0.07)	3.41 (±0.40)
i_{36}	1.11 (±0.10)	13.12 (±1.27)
$q_{g,1}$	0.08 (±0.012)	0.66 (±0.058)
$q_{g,5}$	0.10 (±0.013)	1.79 (±0.12)
Output variables	**UT**	**Nataf-PCE**
σ	0.15 (±0.041)	0.08 (±0.015)
v_3	0.015 (±0.006)	0.002 (±0.0004)
v_4	0.009 (±0.001)	0.0024 (±0.0006)
i_{33}	2.74 (±0.43)	0.60 (±0.13)
i_{36}	2.95 (±0.39)	1.05 (±0.09)
$q_{g,1}$	0.87 (±0.085)	0.11 (±0.014)
$q_{g,5}$	0.31 (±0.038)	0.07 (±0.008)

Note: The values in parentheses indicate the 95% confidence intervals of the RMSPE, calculated from 30 repeated trials with independently sampled input data.

As shown in Table I, the RMSPE values of the GMM-UT algorithm are among the lowest across all approximation methods with nearly all indicators exhibiting the smallest errors below 1%. In contrast, other methods can produce errors exceeding 10% in certain cases. Moreover, the GMM-UT method consistently demonstrates narrower confidence intervals for all variables, indicating not only higher accuracy but also stronger robustness and stability. This suggests that GMM-UT can serve as a reliable alternative to the MC method for solving S-OPF problems. Furthermore, RMSPE is used as a global error metric, and combined with 95% confidence intervals obtained from repeated trials, it further confirms that the Gaussian component reduction process does not significantly distort the tail distributions. The uncertainty in tail estimates is well controlled.

B. Computational Efficiency and Error Analysis

Furthermore, to validate the computational efficiency and reliability of the proposed algorithm, Section IV.B tests the GMM-UT-based S-OPF solving algorithm on Case A and Case B, using the operating point obtained by the MC method as the benchmark. The computational results are shown in Table II, where ε represents the confidence level, k_c represents the number of iterations required for the algorithm to converge, OBJ represents the generation cost corresponding to the final operating point as defined by (5) (The values in parentheses indicate the increase relative to the baseline method, i.e., the MC method), $\bar{t}_{pf}$ represents the average computation time for the power flow calculation in each iteration stage with the values in parentheses indicating the improvement relative to the benchmark method (MC method), η_p and η_v represent RMSPE of the active power output and voltage magnitude of the generators respectively, ε_J represents the maximum probability of all chance constraint violations at the given operating point, with the calculation

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TABLE II
RMSPE OF SYSTEM OPERATION POINTS AT DIFFERENT CONFIDENCE LEVELS

Group 1: Case A - IEEE 14-bus system, ε=5%					
Method	OBJ ($)	$\bar{t}_{pf}$ (s)	ε_J (%)	η_p (%)	η_v (%)
GMM-UT	5378.89(↑ 0.025%)	0.12	5.00	0.004	0.13
UT	5380.04(↑ 0.046%)	0.09	10.07	0.042	0.64
Nataf-PEM	5379.28(↑ 0.032%)	0.52	8.07	0.011	1.04
Nataf-PCE	5378.94(↑ 0.025%)	0.26	5.13	0.003	0.15
Group 2: Case A - IEEE 14-bus system, ε=10%					
Method	OBJ ($)	$\bar{t}_{pf}$ (s)	ε_J (%)	η_p (%)	η_v (%)
GMM-UT	5366.28(↑ 0.188%)	0.12	8.00	0.0020	0.022
UT	5375.85(↑ 0.366%)	0.07	10.27	0.0027	1.029
Nataf-PEM	5376.06(↑ 0.370%)	0.463	9.33	0.0021	0.797
Nataf-PCE	5366.41(↑ 0.189%)	0.20	9.05	0.0007	0.056
Group 3: Case D - Illinois 200-bus system, ε=5%					
Method	OBJ ($)	$\bar{t}_{pf}$ (s)	ε_J (%)	η_p (%)	η_v (%)
GMM-UT	17195.82(↑ 0.03%)	11.50	5.18	0.12	0.003
UT	17250.13(↑ 0.35%)	3.31	8.00	0.31	0.014
Nataf-PEM	17240.81(↑ 0.29%)	32.75	10.27	0.07	0.025
Nataf-PCE	17197.73(↑ 0.04%)	27.16	6.30	0.14	0.027
Group 4: Case D - Illinois 200-bus system, ε=10%					
Method	OBJ ($)	$\bar{t}_{pf}$ (s)	ε_J (%)	η_p (%)	η_v (%)
GMM-UT	16772.60(↑ 0.38%)	11.52	10.77	0.0003	0.003
UT	16910.19(↑ 1.20%)	5.29	18.60	0.8651	1.911
Nataf-PEM	16871.28(↑ 0.97%)	23.98	14.75	0.0624	1.420
Nataf-PCE	16858.73(↑ 0.89%)	24.24	10.97	0.0072	0.018

method as follows:

$$\hat{\varepsilon}_i = \max\left\{\frac{1}{N_s}\sum_{s=1}^{N_s} 1\left\{x_i(\zeta^{(i)}) \notin \psi_i\right\}\right\} \tag{46}$$

where $\hat{\varepsilon}_i$ represents the empirical violation probability of the i-th chance constraint, N_s is the total number of samples of renewable energy, $x_i(\zeta^{(i)})$ is the value of x_i under the i-th output sample, ψ_i is the feasible region of the variable x_i and the indicator function $1\{\cdot\}$ takes the value of 1 if the condition in parentheses holds, otherwise it takes the value of 0.

For the purpose of comparison, no parallel computation is used in any of the cases. It can be observed that, compared to the solution framework based on the MC method, the GMM-UT-based S-OPF solving algorithm proposed in this paper exhibits clear advantages in computational efficiency, objective function error, and constraint error. Although the Nataf-PEM method delivers acceptable accuracy, its computational efficiency remains low. The UT method can be regarded as a special case of the proposed method with only one Gaussian component; while it offers faster computation, it fails to guarantee accuracy, particularly performing poorly in terms of decision error ε_J. In contrast, compared with the Nataf-PCE method, the GMM-UT method often demonstrates more pronounced advantages in both computational speed and OBJ error when solving large-scale systems. However, during the solution process of the S-OPF problem, by observing the RMSPE of the operating points obtained using the GMM-UT method, it can be seen that the solution method based on GMM-UT has certain errors. These errors can mainly be attributed to the following two factors:

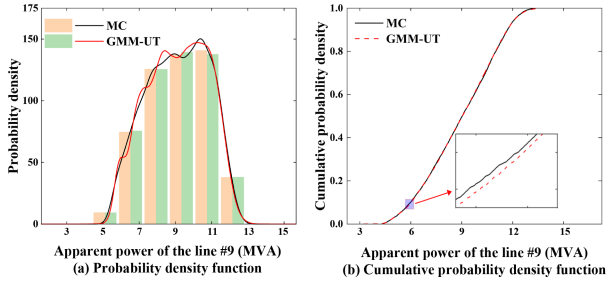


Fig. 7. Comparison of fitting results between GMM-UT and MC algorithms.

TABLE III
TIME CONSUMPTION OF EACH COMPUTATION STEP (UNIT: %)

Test Case	Power Flow Calculations	D-OPF Solve	Other Steps
Case A	94.2	3.3	2.5
Case B	95.4	2.6	2.0
Case C	95.9	3.0	1.2
Case D	96.7	1.9	1.4

Note: OBJ ($) obtained using the MC method for Group 1 to Group 4 are 5377.57, 5356.26, 17190.60 and 16709.33, respectively. $\bar{t}_{pf}$ (s) are 19.67, 19.62, 135.61 and 132.72. The values of ε_J, η_p and η_v are calculated based on the operating points obtained by the MC method, which serve as the benchmark value.

1) During the process of modeling the input random variables using the GMM model, errors are inevitably introduced, leading to a less than perfectly accurate representation of the input variables.

2) Although the GMM model can effectively approximate the actual probability distribution of the output variables, there is still some inherent bias in this fit. As a result, fitting the output variables with the GMM model introduces cumulative errors in each iteration, which impacts the overall solution accuracy. As shown in Fig. 7, taking the scenario $\varepsilon = 10\%$ in Case B as an example, the probability density obtained using the GMM-UT algorithm is generally consistent with that obtained using the MC method. However, the deviation in the fit of the 10th percentile leads to a bias in the final result. Nevertheless, as mentioned earlier, in the vast majority of scenarios set in this study, the method proposed in this chapter achieves accuracy comparable to the MC method. At the same time, compared to other approximation methods, the proposed method not only offers higher accuracy but also maintains satisfactory computational efficiency.

To further analyze the computational cost, Table III presents the runtime distribution of the proposed algorithm for Cases A–D. The total computation time is mainly distributed among three components: power flow calculations under UT sampling, D-OPF solve and other steps including bound tightening. As shown in Table III, the dominant computational burden comes from repeatedly solving AC power flows for each Gaussian component and its associated UT sigma points. In contrast, the computation times for the D-OPF and other steps are relatively minor. This result further confirms the effectiveness of the proposed Gaussian component reduction strategy, which

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TABLE IV
SOLUTION RESULTS BASED ON THE SIMPLIFIED GMM

Group 1: Case C - European 89-bus system, $M_G = 15$, $\varepsilon = 5\%$					
k_c	OBJ (\$)	$\bar{t}_{pf}$ (s)	ε_J (%)	η_p (%)	η_v (%)
6	3716.04	1.04(↓43.38%)	5.12	0.02	0.01
Group 2: Case D - Illinois 200-bus system, $M_G = 25$, $\varepsilon = 5\%$					
k_c	OBJ (\$)	$\bar{t}_{pf}$ (s)	ε_J (%)	η_p (%)	η_v (%)
6	17197.06	5.96(↓48.17%)	5.08	0.04	0.002

accelerates the overall computation by reducing the number of power flow evaluations.

As discussed in Section III-C, without considering the GMM simplification strategy proposed in this paper, the total number of power flow calculations required to reconstruct the output distribution is $(2N+1) \times N_G$. Therefore, the computational time increases roughly linearly with respect to both N and N_G, i.e., $\mathcal{O}(N \cdot N_G)$. This explains the observed phenomenon of computation time increasing rapidly as the system size expands. If the number of uncertainty sources increases further, the GMM simplification algorithm proposed in Section III-D can effectively reduce N_G, thereby mitigating the computational growth. In future work, we plan to test the algorithm on systems with more uncertainty sources and explore dimension-reduction techniques to further improve scalability.

C. Effectiveness Analysis of the Simplified Algorithm

In Section IV-C, to validate the effectiveness of the simplification model proposed in Section III-D, the Gaussian components for Case C and D are reduced to 15 and 25, respectively, with $\varepsilon = 5\%$. The computational results are shown in Table IV, where the values in parentheses represent the changes relative to the use of the un-reduced model.

As shown in Table IV, after reducing the original GMM, the computational efficiency of the algorithm can be significantly improved, while retaining sufficient computational accuracy. This validates the effectiveness of the proposed reduced model in reducing algorithm complexity.

As mentioned in Section III-D, the reduced GMM with M_G components, obtained based on Algorithm 2, provides higher accuracy compared to the model directly fitted with M_G components. To illustrate this, using Case C as an example, the number of components is set to 10, 15 and 20 under each setting, the operating point is fixed to compare the two models as well as the probability distribution obtained using the MC method. The results are shown in Figs. 8, 9 and 10 where GMM-Simplified refers to the reduced model and GMM-Original refers to the directly fitted model.

From Fig. 8 to Fig. 10, it can be seen that the GMM simplification method based on Algorithm 2 achieves higher accuracy in different scenarios. This indirectly suggests that when high modeling accuracy for the input variables is required and the number of Gaussian components needed is large, an effective strategy is to initially fit the model with as many Gaussian components as possible and then use the simplification algorithm to decrease the

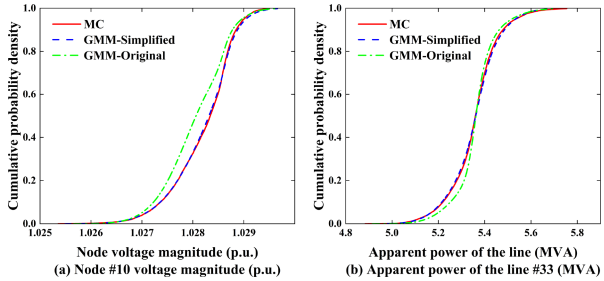


Fig. 8. Comparison of cumulative probability density functions ($M_G = 10$).

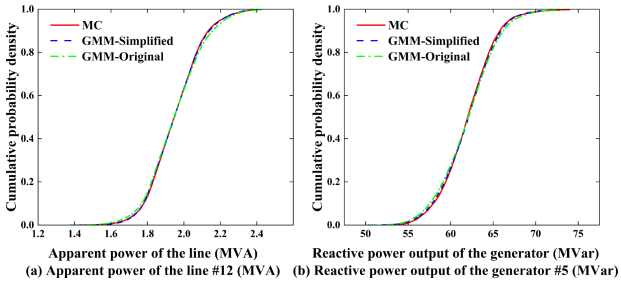


Fig. 9. Comparison of cumulative probability density functions ($M_G = 15$).

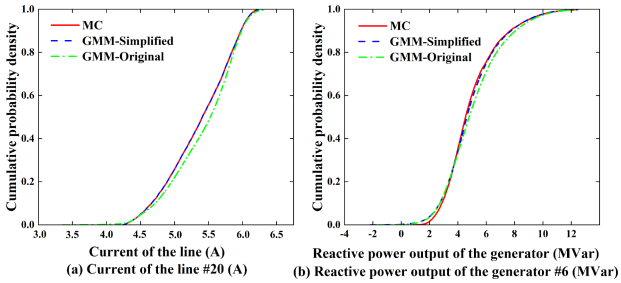


Fig. 10. Comparison of cumulative probability density functions ($M_G = 20$).

number of components. Moreover, the tails of the approximated probability density functions also align well with the benchmark MC results, indicating that the component reduction does not compromise the estimation of low-probability regions critical for chance constraints.

The acceleration proposed in this paper is achieved through two main mechanisms: (1) reducing the number of power flow evaluations from $N_G \times (2N+1)$ to $M_G \times (2N+1)$; (2) adopting a deterministic transformation method based on UT to replace traditional sampling-based approaches. Compared with conventional MC methods which typically require thousands of power flow computations, the proposed framework significantly reduces computational burden while maintaining comparable approximation accuracy.

V. CONCLUSION

This paper aims to address the efficiency challenges encountered in the second stage of the iterative solution process and presents a GMM-UT-based S-OPF solving algorithm as a potential solution. First, an explicit expression model of the S-OPF is constructed. Then, without simplifying the nonlinear power flow model, a fast output variable probability density

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reconstruction method based on the GMM-UT algorithm is proposed to model non-Gaussian input random variables. To solve the issue of reduced GMM-UT algorithm efficiency as the dimensionality of the input random variables increases, a GMM simplification strategy based on KL divergence is further designed. Finally, the proposed algorithm is embedded into a two-stage iterative solving framework, achieving efficient and high-precision solutions for the S-OPF problem. Through the analysis of four different-sized cases, it is verified that the proposed algorithm can obtain the probability distribution of output variables with higher computational efficiency while ensuring sufficient accuracy.

Although the proposed GMM component reduction method achieves significant computational savings while preserving key distributional features, it relies on second-order moment matching and greedy pairwise merging which may not fully capture higher-order statistical properties such as skewness, kurtosis or multimodality. Therefore, future research will focus on enhancing the reduction process to retain higher-order moments and establishing theoretical guarantees on the accuracy and robustness of the simplified GMM in uncertainty propagation tasks. In addition, a complete theoretical analysis of convergence and computational complexity will be pursued in future research.

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